

thus 30-year CMS – 10-year CMS was around $5.25\% - 4.50\% = 75$ bps). So how on earth is this bond able to pay such high coupons in the back years? The answer is that it probably won't. The fact that this bond is callable in three months and every three months thereafter means that it is extremely unlikely that the 25% coupons will ever be paid. Based on the forwards, it is very likely that this bond is called well prior to that time.

Problem Set 7A

Throughout this problem set, use Black's formula to price up options and assume flat skew unless explicitly stated otherwise.

1. You are a derivatives structurer, and one of your clients is interested in purchasing from you a Prime floating rate note. The client specifies that he wants \$100mm of a 5-year note issued by an A-rated corporate. The form of the coupon will be Prime $- C\%$, where Prime is reset daily, averaged, and paid quarterly on an Act/360 basis, and C is a constant that you will determine. Assume that you find an A-rated corporate who is willing to issue at 3-month LIBOR + 15 bps.

You approach your swaps trader who provides you with mid-market swap rates (on a quarterly bond basis) and mid-market levels for Prime-LIBOR basis swaps of various maturities:

Maturity	Quarterly Bond Swap Rate	Prime-LIBOR Basis Swap
1 year	6.00%	-250
2 year	6.25%	-255
3 year	6.50%	-260
4 year	6.75%	-265
5 year	7.00%	-270

For example, your swaps trader is willing to exchange (at mid-market) 3-month LIBOR quarterly money for Prime $- 260$ bps, where Prime is reset daily, averaged, and paid quarterly on an Act/360 basis for a maturity of three years.

a) Assuming that your swaps trader is willing to execute either side of a Prime-LIBOR basis swap 5 bps from mid-market, what level of C can you provide to your client?

b) How much money do you/your swaps trader make in the basis swap? Define profitability as the value of the 5 bps running differential between where the basis swap was executed and mid-market.

For questions 2–5, assume that the swap curve is as follows:

Maturity	Swap Rate
1 year	6.00%
2 year	6.25%
3 year	6.50%
4 year	6.75%
5 year	7.00%

These rates are annual bond versus 1-year LIBOR annual bond.

2. You have a client who is interested in buying a capped LIBOR floating rate note. The coupon will be of the form 1-year LIBOR + y , (coupon) capped at $z\%$. The client wants to purchase \$100mm of a 5-year note issued by an agency. Assume that you find an agency who is willing to fund at 1-year LIBOR – 10 bps for five years. You check with your options trader who informs you that all caps and floors should be priced using a (mid-market) average volatility of 15%.

a) Assume that your client specifies that he would like a coupon of the form 1-year LIBOR + 10 bps, capped at $z\%$. What coupon cap z can you obtain for your client, assuming that all pricing is done at mid-market?

b) Suppose now that your options trader specifies that she will trade caps or floors one vega from mid. That is, the average vol used to price any cap or floor should be moved by one percentage point up or down, depending on whether your options trader is buying or selling the option. Given these transaction costs, what level of z can you provide to your client? What is the profitability on the embedded option?

c) Suppose now that your client specifies that he would like a coupon of the form 1-year LIBOR + y , capped at 7.50%. Assuming mid-market pricing, what level of y can you provide to your client?

d) Now your client is thinking instead of purchasing a *collared floating rate note*. This bond pays a coupon of the form 1-year LIBOR + y , subject to a coupon cap and a coupon floor. Your client specifies the coupon should be of the form 1-year LIBOR + 5 bps, capped at 8.00%. What coupon floor are you able to provide in the note, provided all pricing is done at mid-market? Assume the note matures in five years.

e) Now your client is thinking of purchasing a *super floater*. The client has specified that the leverage factor in the bond should be two. Therefore,

coupons will be of the form $2 \times (1\text{-year LIBOR}) - X\%$, subject to a minimum coupon of 0.00%. Assuming mid-market pricing, what value of X can you provide to your client? Assume the note matures in five years.

3. A client of yours is interested in purchasing a binary accrual note. This is a bond that pays a certain fixed rate if 1-year LIBOR is within a given range and another fixed rate (possibly zero) if LIBOR is outside of the range. That is, the coupons of the bond are:

$$\begin{aligned} &C\% \text{ when LIBOR is within Range} \\ &Y\% \text{ when LIBOR is outside of Range.} \end{aligned}$$

Assume the reset and payment date conventions in the binary accrual note follow the standard swap conventions of “reset in advance and pay in arrears.” What this means in terms of our stylized example is that 1-year LIBOR will be observed at years 0, 1, 2, 3, and 4 and will determine coupon payments for years 1, 2, 3, 4, and 5, respectively. Since 1-year LIBOR today is set, it is known with certainty whether or not the first LIBOR will be in or out of the Range, and so it is also known with certainty if the coupon paid in one year is $C\%$ or $Y\%$. Assume that this bond will be issued by a corporate that funds at LIBOR flat.

a) Suppose your client is confident that 1-year LIBOR will remain between 5.50% and 7.50% over the next five years (note of course that 1-year LIBOR today is 6.00% since the 1-year swap rate is 6.00%). Your client would like you to structure a 5-year binary accrual note that will pay $C\%$ if 1-year LIBOR remains in this range and 0% if LIBOR is outside of this range on any reset date. Assuming that the base vol curve is constant at 15%, what value of C can you provide in the note?

b) Now your client would like to see a variation of the above structure. Your client would like you to structure a 5-year binary accrual note that will pay $C\%$ if 1-year LIBOR remains in this range and 2% if LIBOR is outside of this range on any reset date. Assuming that the base vol curve is constant at 15%, what value of C can you provide in the note?

4. Suppose that you are a derivatives structurer at a dealer. One of your clients has approached you and indicated interest in buying a leveraged inverse floater. The client wants to buy \$100mm of a 5-year bond and wants annual coupons based on 1-year LIBOR. The form of the coupon will be

$$C - 4 \times 1\text{-year LIBOR},$$

where C is a fixed number that you will determine. The client wants to have a minimum coupon guaranteed by the bond of 2%. Assume the issuer is an agency who will be willing to issue in five years at 1-year LIBOR – 25 bps.

Assume that the dealer will trade any cap or floor on 1-year LIBOR using an average vol of 15%.

Thus, to sum things up, you need to structure a trade in which you receive LIBOR – 25, and you pay an inverse floating coupon of the form

$$C - 4 \times \text{LIBOR},$$

subject to a minimum coupon floor of 2%. The goal of this exercise is to determine what is the maximum level of C that you can provide and break even.

a) What type of option will you have to embed in the note to provide the minimum coupon floor? For a \$100mm bond, how much of this option will you buy/sell? Who is long the option? Who is short the option?

b) What is the maximum level of C that you can provide without losing any money?

c) Assume that the moment after you issue the bond, the base vol curve jumps to 16% and nothing else changes. What is the change in the bond value to the investor immediately after the jump in vol? Is this an increase or a decrease in value?

d) For this part, forget part c). Suppose that immediately after the bond is issued, the swap curve has a parallel shift of 50 bps upward (i.e., all rates increase by 50 bps). How does this change affect the value of the underlying swap transaction?

5. In this question you will price up a deleveraged floater, a bond whose coupon depends on a percentage of the movement in LIBOR. Consider a 5-year bond with a coupon of the form

$$0.5 \times \text{LIBOR} + x.$$

Assume that the issuer pays LIBOR flat. What level of x can you provide?

6. Consider the 4-year leveraged inverse floater we priced up in Section 7.7.1. In particular, this is a bond that paid a coupon of

$$\max(16.346\% - 2 \times 1\text{-year LIBOR}, 0). \quad (7.12)$$

Suppose now the investor who bought this bond would like to buy another structured note. The investor proposes that he would like to buy a 4-year note that has some interest rate floors on LIBOR embedded into it and would like a leverage factor of 2 to be used. In particular, the coupons on this bond should be of the form

$$2 \times \max(C\% - 1\text{-year LIBOR}, 0). \quad (7.13)$$

In this question you will solve for the highest strike on the floor, C , that you can provide to the investor.

a) What level of C can you provide to the investor, assuming the same pricing parameters as in Section 7.7.1? That is, assume the swap curve in as in Table 7.5, the base vol curve is flat at 15%, and the issuer funds at LIBOR flat.

b) How does this bond compare to the leveraged inverse floater we structured in Section 7.7.1? That is, how do the coupons in Equation 7.12 compare to the coupons in Equation 7.13? Explain.

Problem Set 7B
Case Study: Target Redemption Note

Table 7.15 shows the terms of a structured note known as a *target redemption note*.

The purpose of this problem is to gain an understanding of the risks inherent in the note from the point of view of the bondholder (and from the point of view of the swap dealer in the underlying swap transaction), as well as to rip apart (i.e., reverse engineer) this note to find the price of some of the embedded exotic options.

The Bond

The salient features of this note are found in Table 7.15.

Historical LIBOR and Scenario Analysis

Figure 7.16 shows a history of 3-month LIBOR for the nearly 10-year period prior to the settlement of this bond. Table 7.16 contains scenario analysis that shows how the bond would perform under a hypothetical sequence of 3-month LIBOR rates.

Note that the level of 3-month LIBOR does not matter whatsoever in the first year; the bondholder will always accrue a coupon of 10% in the first year.

Issuer:	Bank
Principal:	\$100mm
Price:	100% of Principal
Settlement:	September 20, 2003
Maturity:	September 20, 2013, subject to Early Redemption Provision
Coupons:	Year 1: 10% Years 2–10: $10\% - (2 \times 3\text{-month LIBOR})$, subject to a minimum coupon of 0%
Resets and Payment:	Quarterly 30/360
Lifetime Target:	13.05%
Early Redemption Provision:	If the sum of the accrued coupons over the life of the note ever reaches the Lifetime Target, then the bond is redeemed early at par

Table 7.15: Terms of the Target Redemption Note

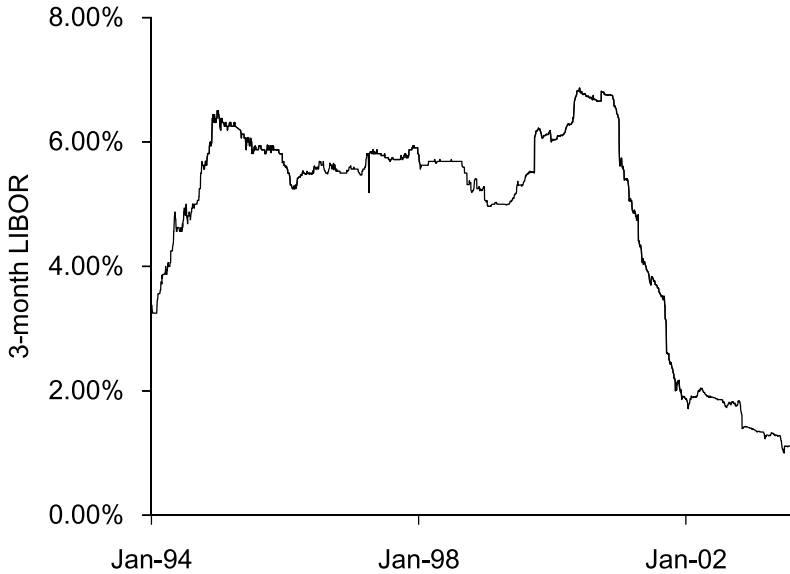


Figure 7.16: Historical 3-Month LIBOR

In the fifth quarter it is assumed that 3-month LIBOR sets at 2.40%. Given this rate setting, the coupon sets at $10\% - 2 \times 3\text{-month LIBOR} = 5.20\%$. This coupon accrues for the quarter on a 30/360 basis, meaning the bondholder receives $5.20\% \times 90/360 = 1.30\%$ for the quarter. Thus the sum of accrued coupons at the end of the fifth quarter is the 10% coupon earned in the first year plus the 1.30% accrual in the fifth quarter for a total of 11.30%. Given the assumed path of 3-month LIBOR, the Lifetime Target of 13.05% is reached in the 7th quarter, at which point in time the bond redeems early and the bondholder receives back his principal.³³

The scenario analysis presented in Table 7.16 shows just one hypothetical path of 3-month LIBOR and associated coupons and timing of principal redemption over the life of the note. When analyzing this note, it is important to consider all possible outcomes that could affect the return to the bondholder.

- a) If you own this bond, what is the best thing that can happen to you?
- b) If you own this bond, what is the worst thing that can happen to you?

³³Note that in the 7th quarter the bondholder only receives a partial coupon so that the sum of accrued coupons received by the bondholder does not exceed the lifetime target.

Quarter	3-month LIBOR	Coupon	Sum of Accrued Coupon
1	not relevant	10.00%	2.50%
2	not relevant	10.00%	5.00%
3	not relevant	10.00%	7.50%
4	not relevant	10.00%	10.00%
5	2.40%	5.20%	11.30%
6	2.60%	4.80%	12.50%
7	2.80%	4.40%	13.05%
			(final coupon 0.55%)

Table 7.16: Scenario Analysis Given a Hypothetical Path of 3-Month LIBOR

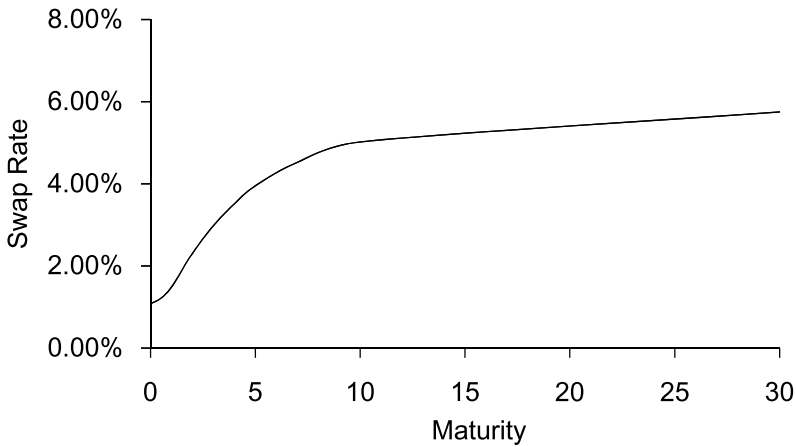


Figure 7.17: The Swap Curve on August 25, 2003

c) What options are embedded into this bond? Can you definitively say whether or not the bondholder is long or short these options, regardless of the prevailing interest rate environment?

Assume that the pricing of this note depended on the swap curve that prevailed on August 25, 2003. The swap curve on this day is shown in Figure 7.17. The level of 3-month LIBOR on August 25, 2003 was 1.14%.

Given the swap curve shown in Figure 7.17, 3-month LIBOR forwards for the 40 quarters between September 20, 2003 and September 20, 2013 have been computed using an industry standard swap model and are provided in Table 7.17.

d) Based on the forwards shown in Table 7.17, provide a scenario analysis like the one provided in Table 7.16. That is, assuming that the *forwards are realized*, what are the projected coupons in the bond and when will the bondholder receive his principal back? We might loosely think of the forwards as the market's expectation of where LIBOR will be in the future. In comparison to these forwards, do you think that the scenario analysis shown in Table 7.16 is a reasonable representation of what may happen if you own this bond?

The Underlying Swap Transaction

This bond is created through an underlying swap transaction between the issuer and a swap dealer. This swap transaction is shown in Figure 7.18.

In the swap, the dealer pays the structured note coupons equal to 10% in the first year and $10\% - 2 \times 3\text{-month LIBOR}$, floored at 0% in years 2 through 10. The issuer is assumed to pay a floating rate of 3-month LIBOR + 55 bps. The swap has a knockout feature: if the sum of accrued coupons over the life of the swap ever exceeds the Lifetime Target of 13.05%, then the swap knocks out. We looked at an example of a knockout swap in Chapter 6. This is another example of a type of knockout swap, albeit a much more complicated one. The knockout feature here is path dependent. This swap will not knock out based on a single observation of LIBOR but rather on the path of LIBOR that is observed over the life of the swap. One of the things that we will be interested in doing is deducing the price that the swap dealer is implicitly attributing to this exotic knockout feature, without having to resort to computing the value of the knockout directly.³⁴

In order to figure out the price of this embedded knockout feature, it will be useful to decompose the transaction in Figure 7.18 in a manner analogous to how we decomposed cancelable swaps. In Figure 6.1, we showed that a cancelable swap can be decomposed into a plain vanilla swap and a receiver swaption. Moreover, since a cancelable swap is an $NPV = 0$

³⁴This is not to suggest that computing the value of the knockout directly is not an important exercise. But the idea here is to come up with a slick way to back out the price